DC Circuits — Reference

prefixes

Symbol	Prefix	Power	Multiply by	In words
G	giga	10^{9}	1,000,000,000	billion
M	mega	10^{6}	1,000,000	million
k	kilo	10^{3}	1,000	thousand
—	(none)	10^{0}	1	one
m	milli	10^{-3}	0.001	thousandth
μ	micro	10^{-6}	0.000 001	millionth
n	nano	10^{-9}	0.000 000 001	billionth
p	pico	10^{-12}	0.000 000 000 001	trillionth

quantities and units

Quantity	Symbol	Unit	In base units
charge	q, Q	coulomb (C)	A·s
current	i, I	ampere (A)	C/s
voltage	v, V	volt (V)	J/C
power	p, P	watt (W)	J/s = V·A
energy	w, W, E	joule (J)	W·s
resistance	R	ohm (Ω)	V/A
conductance	G	siemens (S)	A/V = $1/\Omega$
time	t	second (s)	—

Lowercase (q, i, v, p, w) means the value can change with time. Uppercase (Q, I, V, P, W) means a constant.

constants and conversions

Item	Value
charge of one electron	$e = 1.602 \times 10^{-19}$ C
electrons in 1 C	6.242×10^{18}
minute, hour	1 min = 60 s, 1 h = 3600 s
ampere-hour	1 Ah = 3600 C, 1 mAh = 3.6 C
watt-hour	1 Wh = 3600 J
kilowatt-hour	1 kWh = 3.6×10^{6} J
specific heat of water	$c = 4186$ J/(kg·°C)
mass of water	1 L ≈ 1 kg, 1 fl oz ≈ 0.02957 kg
cents to dollars	divide by 100

formula sheet

1–3. Charge and current

charge → electrons	$n = \dfrac{\lvert Q \rvert}{e}$
electrons → charge	$Q = n e$
constant current	$I = \dfrac{Q}{t}, \quad Q = I t$
constant current, t_1 to t_2	$Q = I(t_2 - t_1)$
changing charge → current	$i = \dfrac{dq}{dt}$
changing current → charge	$q = \int_{t_1}^{t_2} i\,dt$

4. Voltage

voltage	$v = \dfrac{w}{q}$
energy to move a charge	$w = q v$

5–6. Power and energy

power	$p = v i$ (sign: section 5)
power at one instant t_1	$p(t_1) = v(t_1) \cdot i(t_1)$
constant power → energy	$W = P t$
changing power → energy	$w = \int_{t_1}^{t_2} p\,dt$
energy → power	$p = \dfrac{dw}{dt}$
conservation	$\sum p = 0$

7. Batteries and utility energy

power, voltage → current	$I = \dfrac{P}{V}$
Ah rating → Wh rating	$\text{Wh} = V \cdot \text{Ah}$
run time from Ah	$t = \dfrac{\text{Ah}}{I}$ (hours)
run time from Wh	$t = \dfrac{\text{Wh}}{P}$ (hours)
batteries an energy fills	$n = \dfrac{W}{W_{\text{one}}}$, round down

8. Heat and cost

heat	$E = c m \Delta T$
temperature change	$\Delta T = \dfrac{E}{c m}$
cost	cost = kWh · rate

10. Resistors

Ohm's law	$v = i R$
conductance	$G = \dfrac{1}{R}, \quad i = G v$
power in a resistor	$p = i^2 R = \dfrac{v^2}{R} = v i$
energy in a resistor, constant I	$W = I^2 R t$

11–17. Circuit laws

branches, nodes, loops	$b = l + n - 1$
KCL	$\sum i_{\text{in}} = \sum i_{\text{out}}$
KVL	$\sum v = 0$ around a loop
series	$R_{eq} = R_1 + R_2 + \cdots$
parallel	$\dfrac{1}{R_{eq}} = \dfrac{1}{R_1} + \dfrac{1}{R_2} + \cdots$
two in parallel	$R_{eq} = \dfrac{R_1 R_2}{R_1 + R_2}$
voltage divider (series)	$v_1 = \dfrac{R_1}{R_1 + R_2} v$
current divider (parallel)	$i_1 = \dfrac{R_2}{R_1 + R_2} i$
$\Delta \to Y$	$R_a = \dfrac{R_{ab} R_{ca}}{\Sigma}, \quad \Sigma = R_{ab} + R_{bc} + R_{ca}$
$Y \to \Delta$	$R_{ab} = \dfrac{N}{R_c}, \quad N = R_a R_b + R_b R_c + R_c R_a$

calculus tables

Derivatives (a, ω, n, c are constants):

Function	Derivative
c	0
t^n	$n t^{n-1}$
e^{at}	$a e^{at}$
$\sin(\omega t)$	$\omega \cos(\omega t)$
$\cos(\omega t)$	$-\omega \sin(\omega t)$
$f \cdot g$ (product rule)	$f'g + fg'$

Integrals:

Function	Antiderivative
c	ct
t^n	$\dfrac{t^{n+1}}{n+1}$
e^{at}	$\dfrac{e^{at}}{a}$
$\sin(\omega t)$	$-\dfrac{\cos(\omega t)}{\omega}$
$\cos(\omega t)$	$\dfrac{\sin(\omega t)}{\omega}$

Definite integral: $\int_{t_1}^{t_2} f\,dt = F(t_2) - F(t_1)$. F is the antiderivative. Top minus bottom.

Exponent rules:

$e^{at} \cdot e^{bt} = e^{(a+b)t}$	$\left(e^{at}\right)^2 = e^{2at}$
$e^0 = 1$	$e^{-ab} \to 0$ when $b \to \infty$ $(a > 0)$

Shortcuts for a decaying exponential ($a > 0$):

Integral	Result
$\int_0^T Ae^{-at}\,dt$	$\frac{A}{a}\left(1-e^{-aT}\right)$
$\int_0^\infty Ae^{-at}\,dt$	$\frac{A}{a}$
$\int_0^\infty At\,e^{-at}\,dt$	$\frac{A}{a^2}$

Use the shortcuts only to check. Section 3 shows the full steps.

which tool? — given → find

Given → find	Do this	
charge Q → electrons	$n = \lvert Q\rvert/e$	(1)
constant numbers I, t → Q	multiply: $Q = It$	(2)
$q(t)$ → i	differentiate	(3)
$i(t)$ → q	integrate	(3)
w, q → v	divide: $v = w/q$	(4)
v, i → p	multiply, sign from the convention	(5)
$p(t)$ → w	integrate (constant: $W = Pt$)	(6)
$w(t)$ → p	differentiate	(6)
$v(t)$, $i(t)$ → p at t_1	put t_1 in each, then multiply	(6)
$v(t)$, $i(t)$ → total energy	multiply, expand, integrate	(6)
battery rating → run time	$t = \text{Ah}/I$ or Wh/P	(7)
mass, ΔT → heat	$E = cm\Delta T$	(8)
joules → cost	$\div\, 3.6\times10^6$, then $\times$ rate	(8)
two of v, i, R → the third	Ohm's law, one resistor	(10)
resistor → power absorbed	i^2R or v^2/R	(10)
currents at a node → last one	KCL	(12)
voltages in a loop → last one	KVL	(13)
network → R_{eq}	series and parallel	(14)
series, total v → one part	voltage divider	(15)
parallel, total i → one branch	current divider	(16)
not series, not parallel	$\Delta \leftrightarrow Y$	(17)

The number in parentheses is the concept section.

DC Circuits — Concepts

1. charge and electrons

Charge is a property of matter, measured in coulombs (C). An electron carries a negative charge of size $e = 1.602 \times 10^{-19}$ C.
Charge to electrons:

$$n = \frac{|Q|}{e} = \frac{|Q|}{1.602 \times 10^{-19}\ \text{C}}$$

Electrons to charge:

$$Q = ne \quad \text{(write a minus sign for electrons)}$$

Put Q in coulombs before you divide. A prefix left on Q makes the count wrong by a power of ten.

Worked: the number of electrons in 2 mC

1. CONVERT THE PREFIX
$2\ \text{mC} = 2 \times 10^{-3}\ \text{C}$ *(milli = 10^{-3})*

↓

2. WRITE THE FORMULA
$n = \dfrac{|Q|}{e}$

↓

3. SUBSTITUTE
$n = \dfrac{2 \times 10^{-3}}{1.602 \times 10^{-19}}$

↓

4. DIVIDE THE NUMBERS
$\dfrac{2}{1.602} = 1.248$

↓

5. DIVIDE THE POWERS OF TEN
$\dfrac{10^{-3}}{10^{-19}} = 10^{-3-(-19)} = 10^{16}$ *(subtract the exponents)*

↓

6. COMBINE
$n = 1.248 \times 10^{16}$ electrons

Worked: (reverse) the charge of 5×10^{15} electrons

1. WRITE THE FORMULA
$Q = ne$

↓

2. SUBSTITUTE
$Q = \left(5 \times 10^{15}\right)\left(1.602 \times 10^{-19}\right)\ \text{C}$

↓

3. MULTIPLY THE NUMBERS
$5 \cdot 1.602 = 8.01$

↓

4. MULTIPLY THE POWERS OF TEN
$10^{15} \cdot 10^{-19} = 10^{15+(-19)} = 10^{-4}$ *(add the exponents)*

↓

5. COMBINE AND ADD THE SIGN
$Q = -8.01 \times 10^{-4}\ \text{C} = -0.801\ \text{mC}$ *(electrons are negative)*

↓

6. CHECK
$\dfrac{8.01 \times 10^{-4}}{1.602 \times 10^{-19}} = 5 \times 10^{15}$ ✓

2. current: the constant case

Current is the rate of flow of charge, measured in amperes (A). $1\ \text{A} = 1\ \text{C/s}$.
Direct current (DC) is a constant current I.

Current from charge and time: $I = \dfrac{Q}{t}$
Charge from current and time: $Q = It$
Charge over a time range t_1 to t_2: $Q = I(t_2 - t_1)$
Time must be in seconds. A coulomb is an ampere times a second. Convert minutes and hours first.

Conventional current flows the other way from electrons. The current arrow shows the flow of positive charge. A negative current means the real flow goes against the arrow.

Worked: 0.5 A flows for 2 min. Find the charge

1. CONVERT THE TIME
$2\ \text{min} = 2 \cdot 60 = 120\ \text{s}$

↓

2. WRITE THE FORMULA
$Q = It$

↓

3. SUBSTITUTE AND MULTIPLY
$Q = 0.5\ \text{A} \cdot 120\ \text{s} = 60\ \text{C}$

Worked: (reverse) 90 C passes in 30 s. Find the current

1. WRITE THE FORMULA
$I = \dfrac{Q}{t}$

↓

2. SUBSTITUTE AND DIVIDE
$I = \dfrac{90\ \text{C}}{30\ \text{s}} = 3\ \text{A}$

↓

3. CHECK
$Q = It = 3 \cdot 30 = 90\ \text{C}$ ✓

Worked: 4 mA flows from $t = 2$ s to $t = 7$ s. Find the charge

1. LENGTH OF THE RANGE
$t_2 - t_1 = 7 - 2 = 5\ \text{s}$

↓

2. SUBSTITUTE AND MULTIPLY
$Q = 4\ \text{mA} \cdot 5\ \text{s} = 20\ \text{mC}$ *(mA·s = mC, so the prefix passes through)*

3. current: the changing case

If the charge changes with time, $q(t)$, the current is its rate of change. The current is a function too, $i(t)$.
Charge to current (differentiate):

$$i(t) = \frac{dq}{dt}$$

Current to charge (integrate):

$$q = \int_{t_1}^{t_2} i(t)\,dt$$

Which way? — charge or current

Given → find	Do this
$q(t) \to i(t)$	Differentiate $q(t)$.
$i(t) \to q$	Integrate $i(t)$ between the two times.
constant Q, I, t	Section 2 (multiply or divide).

If the problem gives charge and your answer has a higher power of t, you integrated. Go the other way.
Put a time into the function only after you differentiate. A number put into $q(t)$ first becomes a constant, and its derivative is 0.
Units: the prefix passes through. mC → mA. nC → nA. μA → μC.

Worked: (differentiate) $q(t) = 3t^2$ mC. Find $i(t)$

1. PICK THE DIRECTION
Given charge, find current $\Rightarrow$ differentiate.

↓

2. POWER RULE
$\frac{d}{dt}\left(t^n\right) = n\,t^{n-1}$, here $n = 2$

↓

3. APPLY IT
$\frac{d}{dt}\left(3t^2\right) = 3 \cdot 2 \cdot t^{2-1} = 6t$

↓

4. UNITS
$i(t) = 6t$ mA *(mC per second = mA)*

↓

5. CHECK: INTEGRATE BACK
$\int_0^t 6\tau\,d\tau = 3t^2$ mC ✓

Worked: (differentiate) $q(t) = 0.8\left(1 - e^{-0.5t}\right)$ nC. Find $i(t)$

1. DISTRIBUTE
$q(t) = 0.8 - 0.8e^{-0.5t}$ nC

↓

2. DERIVATIVE OF THE CONSTANT
$\frac{d}{dt}(0.8) = 0$

↓

3. DERIVATIVE OF THE EXPONENTIAL
Rule: $\frac{d}{dt}e^{at} = a\,e^{at}$, here $a = -0.5$.
$\frac{d}{dt}\left(-0.8e^{-0.5t}\right) = -0.8 \cdot (-0.5)\,e^{-0.5t} = 0.4e^{-0.5t}$

↓

4. ADD THE TERMS
$i(t) = 0 + 0.4e^{-0.5t} = 0.4e^{-0.5t}$ nA

↓

5. CHECK THE ENDS
$t = 0$: $e^0 = 1$, so $i = 0.4$ nA. $t \to \infty$: $i \to 0$ while q levels off at 0.8 nC. ✓

Worked: (product rule) $q(t) = 0.5t\cos(100\pi t)$ μC. Find $i(t)$

1. NAME THE TWO FACTORS
$f = 0.5t, \quad g = \cos(100\pi t)$

↓

2. DIFFERENTIATE f
$f' = 0.5$

↓

3. DIFFERENTIATE g
Rule: $\frac{d}{dt}\cos(\omega t) = -\omega\sin(\omega t), \quad \omega = 100\pi$.
$g' = -100\pi\sin(100\pi t)$

↓

4. FIRST TERM $f'g$
$0.5\cos(100\pi t)$

↓

5. SECOND TERM fg'
$0.5t \cdot \left(-100\pi\sin(100\pi t)\right) = -50\pi t\sin(100\pi t)$

↓

6. ADD
$i(t) = 0.5\cos(100\pi t) - 50\pi t\sin(100\pi t)$ μA

Worked: (integrate) $i(t) = 20e^{-0.5t}$ mA, from $t = 0$ to $t = 4$ s. Find q

1. SET UP THE INTEGRAL
$q = \int_0^4 20e^{-0.5t}\,dt$

↓

2. ANTIDERIVATIVE
Rule: $\int e^{at}\,dt = \frac{e^{at}}{a}, \quad a = -0.5$.
$F(t) = \frac{20}{-0.5}e^{-0.5t} = -40e^{-0.5t}$

↓

3. EVALUATE AT THE TOP ($t = 4$)
$F(4) = -40e^{-0.5\cdot 4} = -40e^{-2} = -40(0.1353) = -5.413$

↓

4. EVALUATE AT THE BOTTOM ($t = 0$)
$F(0) = -40e^0 = -40(1) = -40$ *($e^0 = 1$, not 0)*

↓

5. TOP MINUS BOTTOM
$q = F(4) - F(0) = -5.413 - (-40) = -5.413 + 40 = 34.59$ mC *(mA·s = mC)*

↓

6. CHECK WITH THE SHORTCUT
$\frac{A}{a}\left(1 - e^{-aT}\right) = \frac{20}{0.5}\left(1 - 0.1353\right) = 40(0.8647) = 34.59$ ✓

Worked: (integrate to ∞) $i(t) = 0.4e^{-0.5t}$ nA. Find the total charge

1. SET UP WITH A FINITE TOP b
$q = \lim_{b\to\infty} \int_0^b 0.4e^{-0.5t}\,dt$ *(you cannot put ∞ into a function, so use b)*

↓

2. ANTIDERIVATIVE
$F(t) = \frac{0.4}{-0.5}e^{-0.5t} = -0.8e^{-0.5t}$

↓

3. EVALUATE AT THE TOP ($t = b$)
$F(b) = -0.8e^{-0.5b}$

↓

4. LET $b \to \infty$
$e^{-0.5b} \to 0$, so $F(b) \to -0.8(0) = 0$

↓

5. EVALUATE AT THE BOTTOM ($t = 0$)
$F(0) = -0.8e^0 = -0.8(1) = -0.8$

↓

6. TOP MINUS BOTTOM
$q = 0 - (-0.8) = 0.8$ nC

↓

7. CHECK WITH THE SHORTCUT
$\frac{A}{a} = \frac{0.4}{0.5} = 0.8$ ✓ *(same as where $q(t)$ levels off two examples up)*

A total to ∞ exists only if $i(t) \to 0$. A constant or a sinusoid has no finite total.

4. voltage

Voltage is the energy needed to move one coulomb of charge between two points. It is measured in volts (V). 1 V $= 1$ J/C.

Voltage from energy and charge: $v = \frac{w}{q}$

Energy from charge and voltage: $w = q\,v$

Current flows *through* an element. Voltage is *across* an element, from its $+$ mark to its $-$ mark. $v_{ab} = -v_{ba}$.

Worked: move 2 C through 12 V. Find the energy

1. WRITE THE FORMULA
$w = q\,v$

↓

2. SUBSTITUTE AND MULTIPLY
$w = 2$ C $\cdot\, 12$ V $= 24$ J *(C·V = C·J/C = J)*

Worked: (reverse) 30 J moves 5 C. Find the voltage

1. WRITE THE FORMULA
$$v = \frac{w}{q}$$

↓

2. SUBSTITUTE AND DIVIDE
$$v = \frac{30\ \text{J}}{5\ \text{C}} = 6\ \text{V}$$

5. power and the passive sign convention

Power is the rate of energy transfer, measured in watts (W). $1\ \text{W} = 1\ \text{J/s}$.

$$p = v\,i$$

Passive sign convention: look at where the current arrow enters the element.
- The arrow enters the $+$ mark: $p = +v\,i$.
- The arrow enters the $-$ mark: $p = -v\,i$.

Then read the sign of the result:
- $p > 0$: the element **absorbs** power (a resistor, a battery that charges).
- $p < 0$: the element **delivers** power (a source).



Conservation of energy: in each circuit, $\sum p = 0$. The power delivered equals the power absorbed.
The sign comes from the arrow and the marks, not from the element type. Check the arrow against the $+$ mark before you multiply.

Worked: absorbs or delivers? Three cases

1. $v = 5$ V, $i = 2$ A **INTO THE $+$ MARK**
Formula: $p = +v\,i$. $p = +(5)(2) = +10$ W.
$p > 0 \Rightarrow$ absorbs 10 W.

↓

2. $v = 5$ V, $i = 2$ A **INTO THE $-$ MARK**
Formula: $p = -v\,i$. $p = -(5)(2) = -10$ W.
$p < 0 \Rightarrow$ delivers 10 W.

↓

3. $v = -3$ V, $i = 4$ A **INTO THE $+$ MARK**
Formula: $p = +v\,i$. $p = +(-3)(4) = -12$ W.
$p < 0 \Rightarrow$ delivers 12 W. *(the negative voltage flips the sign)*

Worked: conservation check

A 12 V source drives 2 A through one resistor. The current leaves the $+$ mark of the source and enters the $+$ mark of the resistor.

1. SOURCE
The arrow enters the $-$ mark. $p = -(12)(2) = -24$ W (delivers)

↓

2. RESISTOR
The arrow enters the $+$ mark. $p = +(12)(2) = +24$ W (absorbs)

↓

3. ADD
$\sum p = -24 + 24 = 0$ ✓

6. energy

Energy is the amount. Power is the rate. The pattern is the same as charge (amount) and current (rate).
Energy from constant power: $W = Pt$ (t in seconds gives joules)
Energy from changing power:

$$w = \int_{t_1}^{t_2} p(t)\,dt = \int_{t_1}^{t_2} v(t)\,i(t)\,dt$$

Power from energy: $p = \dfrac{dw}{dt}$
Power at one instant t_1: $p(t_1) = v(t_1) \cdot i(t_1)$

Which way? — energy or power

Given → find	Do this
constant P, $t \to W$	Multiply: $W = Pt$.
$p(t) \to w$	Integrate $p(t)$ between the two times.
$w(t) \to p$	Differentiate $w(t)$.
$v(t)$, $i(t) \to p(t_1)$	Put t_1 into each. Then multiply.
$v(t)$, $i(t) \to w$	Multiply the functions. Then integrate.

$\int v\,i\,dt$ is not $\left(\int v\,dt\right) \cdot \left(\int i\,dt\right)$. Multiply the functions first.
Units: V·mA = mW. mW·s = mJ.

Worked: (constant) a 12 V battery delivers 2 A for 1 h

1. POWER
$P = V\,I = 12\ \text{V} \cdot 2\ \text{A} = 24\ \text{W}$

↓

2. CONVERT THE TIME
$1\ \text{h} = 3600\ \text{s}$

↓

3. ENERGY
$W = Pt = 24\ \text{W} \cdot 3600\ \text{s} = 86{,}400\ \text{J} = 86.4\ \text{kJ}$

↓

4. SAME ENERGY IN Wh
$24\ \text{W} \cdot 1\ \text{h} = 24\ \text{Wh}$. Check: $24 \cdot 3600 = 86{,}400$ J ✓

Worked: (reverse) $w(t) = 5t^2$ J. Find the power at $t = 3$ s

1. PICK THE DIRECTION
Given energy, find power ⇒ differentiate.

↓

2. DIFFERENTIATE
$$p(t) = \frac{d}{dt}\left(5t^2\right) = 5 \cdot 2 \cdot t = 10t\ \text{W}$$

↓

3. PUT IN THE TIME
$p(3) = 10 \cdot 3 = 30\ \text{W}$

Worked: $v = 50\left(1 - e^{-0.5t}\right)$ V, $i = 4e^{-0.5t}$ mA. Power at $t = 1$ s

1. THE EXPONENTIAL AT $t = 1$
$e^{-0.5 \cdot 1} = e^{-0.5} = 0.6065$

↓

2. VOLTAGE AT $t = 1$
$v(1) = 50(1 - 0.6065) = 50(0.3935) = 19.67\ \text{V}$

↓

3. CURRENT AT $t = 1$
$i(1) = 4(0.6065) = 2.426\ \text{mA}$

↓

4. MULTIPLY
$p(1) = 19.67\ \text{V} \cdot 2.426\ \text{mA} = 47.7\ \text{mW}$ *(V·mA = mW)*

Worked: same device, total energy from 0 to ∞

1. SET UP
$w = \int_0^{\infty} v(t)\, i(t)\, dt = \int_0^{\infty} 50\left(1 - e^{-0.5t}\right) \cdot 4e^{-0.5t}\, dt$

↓

2. MULTIPLY THE CONSTANTS
$50 \cdot 4 = 200$, so $p(t) = 200\left(1 - e^{-0.5t}\right)e^{-0.5t}$ mW

↓

3. DISTRIBUTE $e^{-0.5t}$
$p(t) = 200\left(e^{-0.5t} - e^{-0.5t} \cdot e^{-0.5t}\right)$

↓

4. COMBINE THE EXPONENTS
$e^{-0.5t} \cdot e^{-0.5t} = e^{(-0.5-0.5)t} = e^{-t}$ *(add the exponents)*

↓

5. DISTRIBUTE THE 200
$p(t) = 200e^{-0.5t} - 200e^{-t}$ mW

↓

6. ANTIDERIVATIVE OF THE FIRST TERM
$\int 200e^{-0.5t}\, dt = \dfrac{200}{-0.5}e^{-0.5t} = -400e^{-0.5t}$

↓

7. ANTIDERIVATIVE OF THE SECOND TERM
$\int 200e^{-t}\, dt = \dfrac{200}{-1}e^{-t} = -200e^{-t}$

↓

8. FULL ANTIDERIVATIVE
$F(t) = -400e^{-0.5t} - \left(-200e^{-t}\right) = -400e^{-0.5t} + 200e^{-t}$ *(first minus second, as in step 5)*

↓

9. EVALUATE AT THE TOP ($t = b$)
$F(b) = -400e^{-0.5b} + 200e^{-b}$

↓

10. LET $b \to \infty$
$e^{-0.5b} \to 0$ and $e^{-b} \to 0$, so $F(b) \to -400(0) + 200(0) = 0$

↓

11. EVALUATE AT THE BOTTOM ($t = 0$)
$F(0) = -400e^{0} + 200e^{0} = -400(1) + 200(1) = -200$

↓

12. TOP MINUS BOTTOM
$w = 0 - (-200) = 200$ mJ *(mW·s = mJ)*

↓

13. CHECK WITH THE SHORTCUT $\frac{A}{a}$ ON EACH TERM
$\dfrac{200}{0.5} - \dfrac{200}{1} = 400 - 200 = 200$ ✓

7. battery and utility units: Ah, Wh, kWh

Batteries and utility bills use hours, not seconds.

- **Ampere-hour** (Ah) is charge: 1 Ah $= 1$ A $\cdot\ 3600$ s $= 3600$ C.
- **Watt-hour** (Wh) is energy: 1 Wh $= 1$ W $\cdot\ 3600$ s $= 3600$ J.
- **Kilowatt-hour** (kWh) is energy: 1 kWh $= 1000 \cdot 3600 = 3.6 \times 10^6$ J.

Read the unit of a battery rating. Ah or mAh is charge. Wh or kWh is energy.

Given → find	Do this
P, $V \to I$	$I = \dfrac{P}{V}$
Ah, $V \to$ Wh	Wh $= V \cdot$ Ah
Wh, $V \to$ Ah	Ah $= \dfrac{\text{Wh}}{V}$
Ah rating, $I \to$ run time	$t = \dfrac{\text{Ah}}{I}$ (hours)
Wh rating, $P \to$ run time	$t = \dfrac{\text{Wh}}{P}$ (hours)
Wh rating, V, $I \to$ run time	$P = V\,I$ first, then $t = \dfrac{\text{Wh}}{P}$
Wh → J, Ah → C	multiply by 3600
mAh → Ah	divide by 1000
energy $W \to$ batteries filled	$n = \dfrac{W}{W_{\text{one}}}$, same units, round down

Worked: a 2.5 Ah battery, load draws 500 mA. Run time

1. CONVERT THE CURRENT
500 mA $= 0.5$ A

↓

2. WRITE THE FORMULA
$t = \dfrac{\text{Ah}}{I}$

↓

3. SUBSTITUTE AND DIVIDE
$t = \dfrac{2.5 \text{ Ah}}{0.5 \text{ A}} = 5$ h

Worked: (Ah to Wh) the same battery at 3.7 V

1. WRITE THE FORMULA
Wh $= V \cdot$ Ah

↓

2. SUBSTITUTE AND MULTIPLY
Wh $= 3.7$ V $\cdot\ 2.5$ Ah $= 9.25$ Wh

↓

3. TO JOULES
$9.25 \cdot 3600 = 33{,}300$ J

Worked: a 12 V, 60 Wh battery, load draws 250 mA

1. CONVERT THE CURRENT
250 mA $= 0.25$ A

↓

2. LOAD POWER
$P = V\,I = 12$ V $\cdot\ 0.25$ A $= 3$ W

↓

3. RUN TIME
$t = \dfrac{\text{Wh}}{P} = \dfrac{60 \text{ Wh}}{3 \text{ W}} = 20$ h

↓

4. CHECK WITH CHARGE
Ah $= \dfrac{60}{12} = 5$ Ah. 0.25 A $\cdot\ 20$ h $= 5$ Ah ✓

Worked: a 240 V heater rated 1.2 kW, on for 15 min

1. CONVERT THE POWER
1.2 kW $= 1200$ W

↓

2. CURRENT
$I = \dfrac{P}{V} = \dfrac{1200}{240} = 5$ A

↓

3. CONVERT THE TIME
15 min $= 15 \cdot 60 = 900$ s

↓

4. ENERGY IN JOULES
$W = Pt = 1200 \cdot 900 = 1{,}080{,}000$ J $= 1.08$ MJ

↓

5. SAME ENERGY IN kWh
15 min $= 0.25$ h. 1.2 kW $\cdot\ 0.25$ h $= 0.3$ kWh

↓

6. CHECK
$0.3 \cdot 3.6 \times 10^6 = 1.08 \times 10^6$ J ✓

Worked: how many 3000 mAh, 3.7 V batteries does 1.08 MJ fill?

1. mAh TO Ah
$3000\ \text{mAh} = 3\ \text{Ah}$

↓

2. ENERGY OF ONE BATTERY IN Wh
$3.7\ \text{V} \cdot 3\ \text{Ah} = 11.1\ \text{Wh}$

↓

3. ENERGY OF ONE BATTERY IN J
$11.1 \cdot 3600 = 39{,}960\ \text{J}$

↓

4. DIVIDE
$$n = \frac{1{,}080{,}000}{39{,}960} = 27.03$$

↓

5. ROUND DOWN
27 full charges *(the 0.03 is not a full battery)*

Worked: a pulse of 8 kA for 20 μs across 100 MV

1. CONVERT EACH PREFIX
$I = 8\ \text{kA} = 8 \times 10^{3}\ \text{A}$.
$t = 20\ \mu\text{s} = 20 \times 10^{-6} = 2 \times 10^{-5}\ \text{s}$.
$V = 100\ \text{MV} = 100 \times 10^{6} = 10^{8}\ \text{V}$.

↓

2. CHARGE
$$Q = It = \left(8 \times 10^{3}\right)\left(2 \times 10^{-5}\right) = 16 \times 10^{-2} = 0.16\ \text{C}$$

↓

3. ELECTRONS
$$n = \frac{0.16}{1.602 \times 10^{-19}} = 0.0999 \times 10^{19} \approx 1.0 \times 10^{18}$$

↓

4. POWER
$$P = VI = \left(10^{8}\right)\left(8 \times 10^{3}\right) = 8 \times 10^{11}\ \text{W} = 800\ \text{GW}$$

↓

5. ENERGY
$$W = Pt = \left(8 \times 10^{11}\right)\left(2 \times 10^{-5}\right) = 16 \times 10^{6}\ \text{J} = 16\ \text{MJ}$$

↓

6. CHECK WITH $W = QV$
$0.16 \cdot 10^{8} = 1.6 \times 10^{7}\ \text{J} = 16\ \text{MJ}$ ✓

8. heat and energy cost

Heat to change the temperature of a mass:

$$E = cm\Delta T$$

- c: **specific heat**. Water: $c = 4186\ \text{J/(kg·°C)}$.
- m: mass in kg. Water: $1\ \text{L} \approx 1\ \text{kg}$. $1\ \text{fl oz} \approx 0.02957\ \text{kg}$.
- $\Delta T = T_{\text{final}} - T_{\text{initial}}$, in °C.

Temperature change from energy: $\Delta T = \dfrac{E}{cm}$

If all the electric energy becomes heat: $E_{\text{heat}} = W_{\text{electric}} = VIt$.

Cost: convert joules to kWh, then multiply by the rate.

$$\text{kWh} = \frac{\text{J}}{3.6 \times 10^{6}} \qquad \text{cost} = \text{kWh} \cdot \text{rate}$$

Round down a count of items. 191.1 cups means 191 full cups.

Worked: heat 0.5 kg of water from 20°C to 80°C at 15 cents/kWh

1. TEMPERATURE CHANGE
$\Delta T = 80 - 20 = 60°\text{C}$

↓

2. HEAT
$E = cm\Delta T = 4186 \cdot 0.5 \cdot 60 = 125{,}580\ \text{J}$

↓

3. JOULES TO kWh
$$\frac{125{,}580}{3.6 \times 10^{6}} = 0.03488\ \text{kWh}$$

↓

4. COST OF ONE
$0.03488\ \text{kWh} \cdot 15\ \text{cents/kWh} = 0.523\ \text{cents}$

↓

5. HOW MANY FOR $1
$\$1 = 100$ cents. $\dfrac{100}{0.523} = 191.1 \Rightarrow 191$

Worked: (reverse) 50 kJ goes into 0.4 kg of water. Find ΔT

1. CONVERT THE ENERGY
$50\ \text{kJ} = 50{,}000\ \text{J}$

↓

2. THE BOTTOM: cm
$4186 \cdot 0.4 = 1674.4\ \text{J/°C}$

↓

3. DIVIDE
$$\Delta T = \frac{50{,}000}{1674.4} = 29.9°\text{C}$$

Worked: a heater at 10 A, 120 V for 3 min, same water and rate

1. CONVERT THE TIME
$3\ \text{min} = 3 \cdot 60 = 180\ \text{s}$

↓

2. POWER
$P = VI = 120 \cdot 10 = 1200\ \text{W}$

↓

3. ENERGY
$W = Pt = 1200 \cdot 180 = 216{,}000\ \text{J}$

↓

4. HOW MANY BATCHES
$$\frac{216{,}000}{125{,}580} = 1.72 \Rightarrow 1\ \text{full batch}$$

↓

5. COST
$\dfrac{216{,}000}{3.6 \times 10^{6}} = 0.06\ \text{kWh}$. $0.06 \cdot 15 = 0.9$ cents

9. circuit elements

Active elements can deliver energy (sources). **Passive** elements cannot (resistors, capacitors, inductors).

- **Independent voltage source** (circle): fixed v. The current is whatever the circuit needs.
- **Independent current source** (circle with an arrow): fixed i. The voltage is whatever the circuit needs.
- **Dependent source** (diamond): its value is a gain times a voltage v_x or a current i_x somewhere else in the circuit.

Type	Name	Output	Gain unit
VCVS	voltage-controlled voltage source	$v = \mu v_x$	none
CCVS	current-controlled voltage source	$v = r i_x$	Ω
VCCS	voltage-controlled current source	$i = g v_x$	S
CCCS	current-controlled current source	$i = \beta i_x$	none

Two different ideal voltage sources in parallel, or two different ideal current sources in series, break KVL or KCL.

Worked: the value of a dependent source

1. READ THE TYPE
A diamond labeled $10\,i_x$ with $+$ and $-$ marks: a CCVS. Gain $r = 10\ \Omega$.

↓

2. FIND THE CONTROL
The circuit has $i_x = 0.5$ A.

↓

3. MULTIPLY
$v = 10\,\Omega \cdot 0.5\ \text{A} = 5\ \text{V}$ *($\Omega \cdot A = V$)*

10. Ohm's law and resistor power

Ohm's law:

$$v = iR \qquad i = \frac{v}{R} \qquad R = \frac{v}{i}$$

R is the **resistance** in ohms (Ω). **Conductance** $G = \frac{1}{R}$, in siemens (S). Then $i = Gv$.

Power absorbed by a resistor:

$$p = vi = i^2R = \frac{v^2}{R}$$

Energy absorbed by a resistor: constant current: $W = I^2Rt$. Changing current: $w = \int i^2R\,dt$.

A resistor always absorbs power ($p \geq 0$). All of it becomes heat.

- **Short circuit**: $R = 0$, so $v = 0$ for each current.
- **Open circuit**: $R = \infty$, so $i = 0$ for each voltage.

Ohm's law uses the passive sign convention. If the current arrow enters the $-$ mark, write $v = -iR$.

Which way? — Ohm's law or a definition?

Ohm's law connects v, i, and R of one resistor at the same instant. Use it only when you know two of the three. The definitions ($i = \frac{dq}{dt}$, $p = vi$) work for every element and need no R.

Known → find	Do this
$v, R \to i$	Ohm: $i = \frac{v}{R}$
$i, R \to v$	Ohm: $v = iR$
$v, i \to R$	Ohm: $R = \frac{v}{i}$
$i, R \to p$	$p = i^2R$
$v, R \to p$	$p = \frac{v^2}{R}$
$q(t) \to i$	Definition: $i = \frac{dq}{dt}$. No R.

Worked: 10 V across a 2 kΩ resistor. Find the current

1. CONVERT THE PREFIX
$2\ \text{k}\Omega = 2000\,\Omega$

↓

2. OHM'S LAW
$i = \frac{v}{R} = \frac{10}{2000} = 0.005\ \text{A} = 5\ \text{mA}$

Worked: (reverse) 3 mA through 4.7 kΩ. Find the voltage

1. CONVERT THE PREFIXES
$3\ \text{mA} = 0.003$ A. $4.7\ \text{k}\Omega = 4700\,\Omega$

↓

2. OHM'S LAW
$v = iR = 0.003 \cdot 4700 = 14.1$ V

Worked: power absorbed by a $50\,\Omega$ resistor that carries 0.2 A

1. WAY 1: $p = i^2R$
$i^2 = (0.2)^2 = 0.04$. $p = 0.04 \cdot 50 = 2$ W

↓

2. FIND v FOR THE OTHER WAYS
$v = iR = 0.2 \cdot 50 = 10$ V

↓

3. WAY 2: $p = \frac{v^2}{R}$
$v^2 = 10^2 = 100$. $p = \frac{100}{50} = 2$ W ✓

↓

4. WAY 3: $p = vi$
$p = 10 \cdot 0.2 = 2$ W ✓ *(all three ways agree)*

Worked: energy absorbed by $100\,\Omega$ carrying 0.5 A for 10 min

1. POWER
$P = I^2R = (0.5)^2 \cdot 100 = 0.25 \cdot 100 = 25$ W

↓

2. CONVERT THE TIME
$10\ \text{min} = 10 \cdot 60 = 600$ s

↓

3. ENERGY
$W = Pt = 25 \cdot 600 = 15{,}000\ \text{J} = 15\ \text{kJ}$

Worked: energy absorbed by $10\,\Omega$ with $i(t) = 2e^{-t}$ A, $0 \to \infty$

1. POWER FORMULA
$p(t) = i(t)^2\,R$

↓

2. SQUARE THE CURRENT
$\left(2e^{-t}\right)^2 = 2^2 \cdot \left(e^{-t}\right)^2 = 4e^{-2t}$ *(square each factor, double the exponent)*

↓

3. MULTIPLY BY R
$p(t) = 4e^{-2t} \cdot 10 = 40e^{-2t}$ W

↓

4. SET UP THE INTEGRAL
$w = \lim_{b\to\infty} \int_0^b 40e^{-2t}\,dt$

↓

5. ANTIDERIVATIVE
$F(t) = \frac{40}{-2}e^{-2t} = -20e^{-2t}$

↓

6. EVALUATE AT THE TOP ($t = b$)
$F(b) = -20e^{-2b}$

↓

7. LET $b \to \infty$
$e^{-2b} \to 0$, so $F(b) \to 0$

↓

8. EVALUATE AT THE BOTTOM ($t = 0$)
$F(0) = -20e^0 = -20$

↓

9. TOP MINUS BOTTOM
$w = 0 - (-20) = 20$ J

↓

10. CHECK WITH THE SHORTCUT
$\frac{A}{a} = \frac{40}{2} = 20$ ✓

Worked: (definition, then Ohm) $q(t) = 3t^2$ mC through 2 kΩ. v and p at $t = 1$ s

1. DEFINITION: CHARGE TO CURRENT
$i(t) = \dfrac{dq}{dt} = 6t$ mA. $i(1) = 6$ mA $= 0.006$ A

↓

2. OHM: CURRENT TO VOLTAGE
$v = iR = 0.006 \cdot 2000 = 12$ V

↓

3. POWER ABSORBED
$p = i^2R = (0.006)^2 \cdot 2000 = 0.000036 \cdot 2000 = 0.072$ W $= 72$ mW

↓

4. CHECK
$p = vi = 12 \cdot 0.006 = 0.072$ W ✓

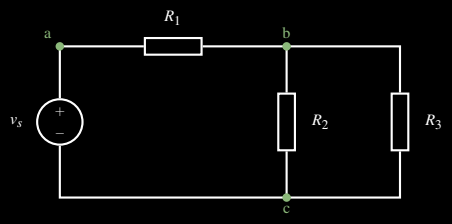
11. nodes, branches, loops

- **Branch**: one element (a source or a resistor).
- **Node**: a point where two or more branches connect.
- **Loop**: a closed path that passes each node one time only.

For b branches, n nodes, and l independent loops: $b = l + n - 1$.

Series: elements that carry the same current. **Parallel**: elements across the same two nodes, so they have the same voltage.

All points joined by plain wire are one node. Merge them before you count.



Worked: count the circuit above

1. BRANCHES
v_s, R_1, R_2, R_3 $\Rightarrow$ $b = 4$

↓

2. NODES
a, b, c *(the top wire from R_2 to R_3 is one node b)* $\Rightarrow$ $n = 3$

↓

3. LOOPS FROM THE FORMULA
$l = b - n + 1 = 4 - 3 + 1 = 2$

↓

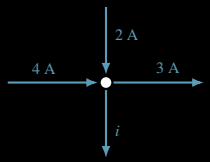
4. CHECK BY LOOKING
Loop 1: v_s, R_1, R_2. Loop 2: R_2, R_3. ✓
R_2 and R_3 share nodes b and c: they are in parallel.

12. Kirchhoff's current law (KCL)

Charge cannot collect at a node. So the current in equals the current out.

$$\sum i_{\text{in}} = \sum i_{\text{out}} \qquad \text{or} \qquad \sum_{k=1}^{N} i_k = 0$$

Sign rule for the second form: into the node is $+$, out of the node is $-$.



Worked: find i in the figure

1. LIST THE CURRENTS IN
4 A and 2 A

↓

2. LIST THE CURRENTS OUT
3 A and i

↓

3. IN EQUALS OUT
$4 + 2 = 3 + i$

↓

4. SOLVE
$6 = 3 + i \Rightarrow i = 6 - 3 = 3$ A

Worked: (negative answer) 1 A in; 3 A and i out

1. IN EQUALS OUT
$1 = 3 + i$

↓

2. SOLVE
$i = 1 - 3 = -2$ A

↓

3. READ THE SIGN
i is negative, so the real current is 2 A into the node, against the arrow.

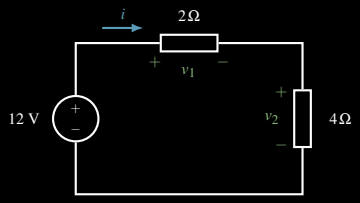
13. Kirchhoff's voltage law (KVL)

Energy is conserved, so the voltages around a closed loop add to zero.

$$\sum v = 0 \qquad \text{or} \qquad \sum v_{\text{rises}} = \sum v_{\text{drops}}$$

Sign rule: walk around the loop in one direction. If you enter an element at its $+$ mark, write $+v$. If you enter at its $-$ mark, write $-v$.

Pick one direction and keep it for the full loop.



Worked: the loop above. Find i, each voltage, and each power

1. WALK CLOCKWISE FROM THE BOTTOM LEFT
Source: enter at $-$ $\Rightarrow -12$. R_1: enter at $+$ $\Rightarrow +v_1$. R_2: enter at $+$ $\Rightarrow +v_2$.

↓

2. KVL
$-12 + v_1 + v_2 = 0$

↓

3. OHM'S LAW FOR EACH RESISTOR
$v_1 = 2i, \quad v_2 = 4i$

↓

4. SUBSTITUTE
$-12 + 2i + 4i = 0$

↓

5. COMBINE LIKE TERMS
$-12 + 6i = 0$

↓

6. SOLVE FOR i
$6i = 12 \Rightarrow i = \dfrac{12}{6} = 2$ A

↓

7. EACH VOLTAGE
$v_1 = 2 \cdot 2 = 4$ V, $\quad v_2 = 4 \cdot 2 = 8$ V

↓

8. POWER ABSORBED BY EACH RESISTOR
$p_1 = i^2 R_1 = 2^2 \cdot 2 = 8$ W, $\quad p_2 = i^2 R_2 = 2^2 \cdot 4 = 16$ W

↓

9. POWER OF THE SOURCE
The arrow enters the $-$ mark. $p = -vi = -(12)(2) = -24$ W (delivers 24 W)

↓

10. CHECK
KVL: $4 + 8 = 12$ ✓ Power: $-24 + 8 + 16 = 0$ ✓

14. series and parallel resistors

Series: one current flows through each resistor. The resistances add.

$$R_{eq} = R_1 + R_2 + \cdots + R_N$$

Parallel: one voltage is across each resistor. The conductances add.

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_N}$$

Two in parallel (product over sum): $R_1 \parallel R_2 = \dfrac{R_1 R_2}{R_1 + R_2}$

N equal resistors R in parallel: $R_{eq} = \dfrac{R}{N}$

A parallel R_{eq} is smaller than the smallest resistor.

The reciprocal formula gives $\frac{1}{R_{eq}}$. Flip the result at the end.

Worked: three in parallel: 10, 20, $20\,\Omega$

1. EACH RECIPROCAL
$\dfrac{1}{10} = 0.1, \quad \dfrac{1}{20} = 0.05, \quad \dfrac{1}{20} = 0.05$

↓

2. ADD
$\dfrac{1}{R_{eq}} = 0.1 + 0.05 + 0.05 = 0.2$ S

↓

3. FLIP
$R_{eq} = \dfrac{1}{0.2} = 5\,\Omega$ *(smaller than 10 ✓)*

Worked: four $100\,\Omega$ ***resistors in parallel***

1. EQUAL-RESISTOR RULE
$R_{eq} = \dfrac{R}{N} = \dfrac{100}{4} = 25\,\Omega$

↓

2. CHECK WITH RECIPROCALS
$4 \cdot \dfrac{1}{100} = 0.04. \quad \dfrac{1}{0.04} = 25\,\Omega$ ✓

Worked: $R_{eq} = 2 + \left[12 \parallel (4 + 6 \parallel 3)\right]\ \Omega$

Start at the part farthest from the terminals. Work toward the terminals.

1. INNERMOST PARALLEL PAIR $6 \parallel 3$
$\dfrac{6 \cdot 3}{6 + 3} = \dfrac{18}{9} = 2\,\Omega$

↓

2. SERIES $4 + 2$
$4 + 2 = 6\,\Omega$

↓

3. PARALLEL $12 \parallel 6$
$\dfrac{12 \cdot 6}{12 + 6} = \dfrac{72}{18} = 4\,\Omega$

↓

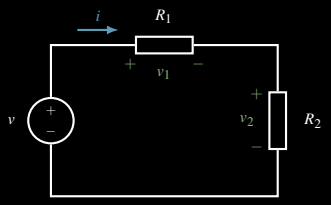
4. SERIES $2 + 4$
$R_{eq} = 2 + 4 = 6\,\Omega$

15. voltage division

Series resistors share the source voltage in proportion to their resistance:

$$v_1 = \frac{R_1}{R_1 + R_2}\, v \qquad v_2 = \frac{R_2}{R_1 + R_2}\, v$$

Own resistor on top. The larger resistor gets the larger voltage.



Worked: 12 ***V across*** $R_1 = 2$ ***kΩ and*** $R_2 = 4$ ***kΩ***

1. TOTAL RESISTANCE
$R_1 + R_2 = 2 + 4 = 6$ kΩ

↓

2. VOLTAGE ON R_1
$v_1 = \dfrac{2}{6} \cdot 12 = 4$ V

↓

3. VOLTAGE ON R_2
$v_2 = \dfrac{4}{6} \cdot 12 = 8$ V

↓

4. CURRENT
$i = \dfrac{12\text{ V}}{6000\,\Omega} = 0.002$ A $= 2$ mA

↓

5. POWER ABSORBED BY EACH
$p_1 = (0.002)^2 \cdot 2000 = 0.008$ W $= 8$ mW
$p_2 = (0.002)^2 \cdot 4000 = 0.016$ W $= 16$ mW

↓

6. CHECK
$4 + 8 = 12$ V ✓ Source: $12 \cdot 0.002 = 24$ mW $= 8 + 16$ ✓

Worked: (reverse)** 12 **V source,** $R_1 = 9$ **kΩ. Find** R_2 **so that** $v_2 = 3$ **V

1. DIVIDER WITH R_2 UNKNOWN
$3 = \dfrac{R_2}{9+R_2} \cdot 12$ *(kΩ throughout)*

↓

2. CLEAR THE FRACTION
$3(9+R_2) = 12R_2$

↓

3. DISTRIBUTE
$27 + 3R_2 = 12R_2$

↓

4. SOLVE
$27 = 9R_2 \Rightarrow R_2 = 3$ kΩ

↓

5. CHECK
$\dfrac{3}{9+3} \cdot 12 = \dfrac{3}{12} \cdot 12 = 3$ V ✓

16. current division

Parallel resistors share the total current i:

$$i_1 = \frac{R_2}{R_1+R_2}\, i \qquad i_2 = \frac{R_1}{R_1+R_2}\, i$$

The *other* resistor goes on top. The smaller resistor gets the larger current. **Which divider?** Series resistors share one current, so the voltage splits (own R on top). Parallel resistors share one voltage, so the current splits (other R on top).

***Worked:** 6 **A into** 3 Ω ∥ 6 Ω*

1. SUM OF THE TWO
$3 + 6 = 9\,\Omega$

↓

2. CURRENT IN THE 3 Ω
$i_1 = \dfrac{6}{9} \cdot 6 = 4$ A *(the other resistor, 6, on top)*

↓

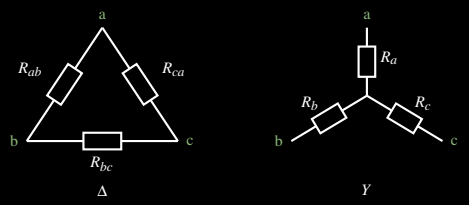
3. CURRENT IN THE 6 Ω
$i_2 = \dfrac{3}{9} \cdot 6 = 2$ A

↓

4. CHECK
KCL: $4 + 2 = 6$ ✓ Same voltage: $4 \cdot 3 = 2 \cdot 6 = 12$ V ✓

17. wye–delta conversion

Some networks are neither series nor parallel (a bridge, for example). Convert a Δ (triangle) to a Y (star), or the reverse. Both connect the same three nodes a, b, c.



$\Delta \to Y$: each Y resistor $= \dfrac{\text{product of the two } \Delta \text{ sides that touch its node}}{\text{sum of all three } \Delta \text{ sides}}$

$$R_a = \frac{R_{ab}R_{ca}}{\Sigma} \qquad R_b = \frac{R_{ab}R_{bc}}{\Sigma} \qquad R_c = \frac{R_{bc}R_{ca}}{\Sigma}$$

$Y \to \Delta$: each Δ side $= \dfrac{N}{\text{the } Y \text{ resistor opposite it}}$, where $N = R_aR_b + R_bR_c + R_cR_a$.

$$R_{ab} = \frac{N}{R_c} \qquad R_{bc} = \frac{N}{R_a} \qquad R_{ca} = \frac{N}{R_b}$$

Balanced: all three equal. Then $R_Y = \frac{R_\Delta}{3}$ and $R_\Delta = 3R_Y$.

***Worked:** Δ → Y **with** $R_{ab} = 10$, $R_{bc} = 20$, $R_{ca} = 30\,\Omega$*

1. SUM
$\Sigma = 10 + 20 + 30 = 60\,\Omega$

↓

2. R_a (TOUCHES R_{ab} AND R_{ca})
$R_a = \dfrac{10 \cdot 30}{60} = \dfrac{300}{60} = 5\,\Omega$

↓

3. R_b (TOUCHES R_{ab} AND R_{bc})
$R_b = \dfrac{10 \cdot 20}{60} = \dfrac{200}{60} = 3.33\,\Omega$

↓

4. R_c (TOUCHES R_{bc} AND R_{ca})
$R_c = \dfrac{20 \cdot 30}{60} = \dfrac{600}{60} = 10\,\Omega$

***Worked: (reverse)** Y → Δ **with** $R_a = 5$, $R_b = 3.33$, $R_c = 10\,\Omega$*

1. THE THREE PRODUCTS
$R_aR_b = 5 \cdot 3.33 = 16.67$
$R_bR_c = 3.33 \cdot 10 = 33.33$
$R_cR_a = 10 \cdot 5 = 50$

↓

2. ADD THEM
$N = 16.67 + 33.33 + 50 = 100$

↓

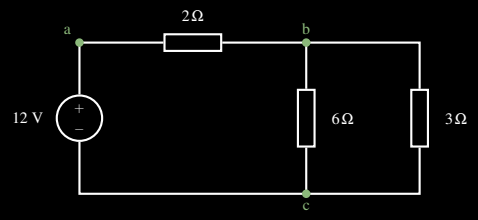
3. EACH SIDE OVER THE OPPOSITE Y
$R_{ab} = \dfrac{100}{R_c} = \dfrac{100}{10} = 10\,\Omega$
$R_{bc} = \dfrac{100}{R_a} = \dfrac{100}{5} = 20\,\Omega$
$R_{ca} = \dfrac{100}{R_b} = \dfrac{100}{3.33} = 30\,\Omega$

↓

4. CHECK
The same 10, 20, 30 as the example above ✓

18. putting it together: one full circuit

This uses sections 10 to 16 in order: reduce, find the source current, then expand back out.



Worked: find every current, voltage, and power

1. PARALLEL PAIR $R_2 \parallel R_3$

$$\frac{6 \cdot 3}{6+3} = \frac{18}{9} = 2\,\Omega$$

↓

2. SERIES WITH R_1

$$R_{eq} = 2+2 = 4\,\Omega$$

↓

3. SOURCE CURRENT (OHM ON R_{eq})

$$i = \frac{12}{4} = 3\text{ A} \quad \textit{(this current flows through } R_1\textit{)}$$

↓

4. VOLTAGE ON R_1

$$v_1 = iR_1 = 3 \cdot 2 = 6\text{ V}$$

↓

5. VOLTAGE ON THE PARALLEL PAIR (KVL)

$$v_{bc} = 12 - 6 = 6\text{ V} \quad \textit{(same on } R_2 \textit{ and } R_3\textit{)}$$

↓

6. CURRENT IN EACH BRANCH (OHM)

$$i_2 = \frac{6}{6} = 1\text{ A}, \quad i_3 = \frac{6}{3} = 2\text{ A}$$

↓

7. KCL CHECK AT NODE b

in 3 A = out $1+2$ A ✓

↓

8. POWER ABSORBED BY EACH RESISTOR

$$p_1 = 3^2 \cdot 2 = 18\text{ W}, \quad p_2 = 1^2 \cdot 6 = 6\text{ W}, \quad p_3 = 2^2 \cdot 3 = 12\text{ W}$$

↓

9. POWER CHECK

Source delivers $12 \cdot 3 = 36$ W. Absorbed: $18+6+12 = 36$ W ✓